

Problem

A 2000-kW turbine-generator of 0.85 power factor operates at the rated load. An additional load of 300 kW at 0.8 power factor is added. What kVAR of capacitors is required to operate the turbine-generator but keep it from being overloaded?

Step-by-step solution

Step 1 of 2

Original Load:-

$$P_1 = 2000 \text{ KW} , \quad \cos \theta_1 = 0.85 \Rightarrow \theta_1 = 31.79^\circ$$

$$S_1 = \frac{P_1}{\cos \theta_1} = 2352.94 \text{ KVA}$$

$$Q_1 = S_1 \sin \theta_1 = 1239.5 \text{ KVAR}$$

Additional Load:-

$$P_2 = 300 \text{ KW} , \quad \cos \theta_2 = 0.8 \Rightarrow \theta_2 = 36.87^\circ$$

$$S_2 = \frac{P_2}{\cos \theta_2} = 375 \text{ KVA}$$

$$Q_2 = S_2 \sin \theta_2 = 225 \text{ KVAR}$$

Total Load:-

$$S = S_1 + S_2 = (P_1 + P_2) + j(Q_1 + Q_2) = P + jQ$$

$$P = 2000 + 300 = 2300 \text{ KW}$$

$$Q = 1239.5 + 225$$

$$= 1464.5 \text{ KVAR}$$

Step 2 of 2

The minimum operating P f for a 2300 K W load and not exceeding the K V A rating of the generator is,

$$\cos \theta = \frac{P}{S_1} = \frac{2300}{2352.94} = 0.9775$$

The maximum load K V A R for this conditions,

$$Q_m = S_1 \sin \theta = 2352.94 \sin(12.177^\circ)$$

$$Q_m = 496.313 \text{ KVAR}$$

The capacitor must supply the difference between the total load

K V A R (i.e.Q) and the permissible generator K V A R (i.e . Qm).

$$\text{Thus, } Q_c = Q - Q_m$$

$$= 968.2 \text{ KVAR} .$$